Machine Learning
Quiz Refresher: Chapter 4, Answer Sheet

Answers to the 20 recall questions, the 8 multiple-choice questions and the 5 exam-style problems

Machine Learning

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Answers

Round A

A1. Discriminative: model $p(y \mid x)$ directly through scores. Generative: model $p(y)$ and $p(x \mid y)$, then apply Bayes' rule.

A2. $\sigma(s) = 1/(1+e^{-s})$; $\sigma(-s) = 1 - \sigma(s)$; boundary at $s = 0$ ($\sigma = \frac{1}{2}$).

A3. $\sigma(s) > t \iff s > \log \frac{t}{1-t}$: the boundary shifts to $s = \log \frac{t}{1-t}$ (negative if $t < \frac{1}{2}$).

A4. $\text{softmax}(s)_c = e^{s_c} / \sum_k e^{s_k}$. The constant multiplies numerator and denominator by the same e^a: only $C-1$ contrasts matter. For $C = 2$ it is $\sigma(s_1 - s_2)$.

A5. $\ell_{\text{CE}}^{(i)} = -\sum_c \mathbf{1}(y^{(i)} = c) \log p_\theta(c \mid x^{(i)}) = -\log p_\theta(y^{(i)} \mid x^{(i)})$; $-\log L_{\text{cond}} = -\sum_i \log p_\theta(y^{(i)} \mid x^{(i)}) = N \widehat{R}_{\text{CE}}$. Minimising cross-entropy is conditional maximum likelihood.

A6. Hyperplanes: $\{x : s_c(x) = s_{c'}(x)\}$ is affine. Costs shift the threshold, hence translate the hyperplane, but never bend it.

A7. $p(x)$ (no model of the inputs); a familiarity or outlier signal; an easy way to adapt to new class priors; the ability to generate inputs.

A8. $Y \sim \text{Categorical}(\pi)$, then $X \mid Y = c \sim p_\theta(\cdot \mid y = c)$; joint $p_\phi(x, y = c) = \pi_c \, p_\theta(x \mid y = c)$.

A9. $p_\phi(c \mid x) = \pi_c p_\theta(x \mid c) / \sum_k \pi_k p_\theta(x \mid k) = \text{softmax}(\log \pi_c + \log p_\theta(x \mid c))_c$; the score is $\log \pi_c + \log p_\theta(x \mid c)$.

A10. $\log p_\phi(x^{(i)}, y^{(i)}) = \log \pi_{y^{(i)}} + \log p_\theta(x^{(i)} \mid y^{(i)})$, and each class has its own θ_c, so the sum separates into a prior part and one term per class over that class's examples. $\hat{\pi}_c = N_c / N$.

A11. $\hat{\mu}_c = \frac{1}{N_c} \sum_{i:y^{(i)}=c} x^{(i)}$; $\widehat{\Sigma}_c = \frac{1}{N_c} \sum_{i:y^{(i)}=c} (x^{(i)} - \hat{\mu}_c)(x^{(i)} - \hat{\mu}_c)^\top$; $d(d+1)/2$ parameters.

A12. QDA: one covariance per class; LDA: one shared covariance. In LDA the quadratic term $-\frac{1}{2} x^\top \widehat{\Sigma}^{-1} x$ is the same for all classes and cancels, leaving affine scores: hyperplane boundaries. QDA keeps it: quadric boundaries.

A13. $p_\theta(x \mid y = c) = \prod_{j=1}^d p_\theta(x_j \mid y = c)$: features conditionally independent given the class. Score $s_c(x) = \log \hat{\pi}_c + \sum_j \log p_{\hat{\theta}}(x_j \mid y = c)$: additive evidence.

A14. $\hat{\theta}_{cj} = (n_{cj} + \alpha) / (\sum_\ell n_{c\ell} + \alpha V)$. With $\alpha = 0$ an unseen word gives $\hat{\theta}_{cj} = 0$, $\log 0 = -\infty$: a single word vetoes the class forever.

A15. Add λs_j^2 (a fraction of the feature's overall variance) to every $\hat{\sigma}_{cj}^2$. It prevents a near-zero class variance from turning one feature into a spike that dominates every score.

A16. Generative wins with little data when its structure (Gaussian, independence) is credible: fewer effective parameters, lower variance. Discriminative wins with much data: fewer assumptions, flexible boundaries. Misspecification is persistent: more data does not fix a wrong $p(x \mid y)$, the generative model converges to the wrong boundary.

A17. The minority class is estimated from few examples, so its $p(x \mid y)$ or its region of the posterior has high variance; the decision part (costs, threshold) is unchanged by imbalance.

A18. Generative: keep $p_{\hat{\theta}}(x \mid y)$, swap in the deployment priors in Bayes' rule. Discriminative: multiply the odds by $\frac{\pi_d/(1-\pi_d)}{\pi_t/(1-\pi_t)}$; it requires r_t to be calibrated.

A19. Softmax outputs compare the known classes with each other and can be near 1 on an input far from all training data. Remedy: a likelihood-based reject option, $\hat{y} = \text{reject}$ if $\log p_{\hat{\theta}}(x \mid c^*) < \tau_{c^*}$ (or any density/outlier score).

A20. Generative pretraining (next-token prediction learns $p(x)$ and a representation) followed by a discriminative head or fine-tuning for the target labels.

Round B

B1. **(c)** $e^{-2.2} = 0.111$, $1/1.111 = 0.900$.

B2. **(a)** $s > \ln \frac{0.2}{0.8} = \ln 0.25 = -1.386$: a cheap false alarm lowers the bar.

B3. **(b)** $e^3/(e + e^2 + e^3) = 20.09/30.19 = 0.665$.

B4. **(d)** $3 \times d(d+1)/2 = 3 \times 55 = 165$. LDA would spend 55.

B5. **(a)** $(0+1)/(9{,}000 + 1 \cdot 1{,}000) = 1/10{,}000 = 10^{-4}$.

B6. **(c)** $\text{odds}_t = 1$; ratio $= (0.1/0.9)/(0.5/0.5) = 1/9$; $\text{odds}_d = 1/9$, so $r_d = (1/9)/(1 + 1/9) = 0.1$.

B7. **(d)** with few examples a credible generative structure lowers variance; with many examples the flexible discriminative model wins, and no model is immune to misspecification.

B8. **(b)** the model never learned $p(x)$; it only learned to separate cats from dogs, and a toaster must still land somewhere on the $(0, 1)$ scale.

Round C

C1. A logistic classifier by hand

(a) $s(x^{(1)}) = 3 - 2 + 0.5 = 1.5$, $\hat{p} = \sigma(1.5) = 1/(1+0.223) = 0.818$. $s(x^{(2)}) = 0 - 2 + 0.5 = -1.5$, $\hat{p} = \sigma(-1.5) = 0.182$.

(b) $\ell^{(1)} = -\ln 0.818 = 0.201$; $\ell^{(2)} = -\ln 0.182 = 1.704$; $\widehat{R}_{\text{CE}} = (0.201 + 1.704)/2 = 0.953$. The second example: its gradient w.r.t. the score is $\hat{p} - y = 0.182 - 1 = -0.818$ against -0.182 for the first. Cross-entropy pushes hardest on confident mistakes.

(c) $1.5x_1 - 2x_2 + 0.5 = 0$, a line. $x^{(1)}$: $s = 1.5 > 0 \Rightarrow$ class 1. $x^{(2)}$: $s = -1.5 < 0 \Rightarrow$ class 0 (wrong).

(d) $t = C_{\text{FP}}/(C_{\text{FP}} + C_{\text{FN}}) = 1/5 = 0.2$; boundary $s > \ln(0.2/0.8) = -1.386$: the line $1.5x_1 - 2x_2 + 0.5 = -1.386$, i.e. $1.5x_1 - 2x_2 + 1.886 = 0$. $x^{(1)}$: still class 1. $x^{(2)}$: $-1.5 < -1.386$, still class 0, but only just. The boundary translated; it is still a parallel line.

(e) $L_{\text{cond}}(\theta) = \prod_i p_\theta(y^{(i)} \mid x^{(i)})$ (independent examples), so $-\log L_{\text{cond}} = -\sum_i \log p_\theta(y^{(i)} \mid x^{(i)}) = N\widehat{R}_{\text{CE}}$. Minimising cross-entropy maximises the conditional likelihood of the labels given the inputs: nothing about $p(x)$ is modelled.

C2. One-dimensional Gaussian classes: LDA versus QDA

(a) $\hat{\pi}_0 = 3/5 = 0.6$, $\hat{\pi}_1 = 0.4$. $\hat{\mu}_0 = 2$, $\hat{\sigma}_0^2 = (1+0+1)/3 = 2/3$. $\hat{\mu}_1 = 6$, $\hat{\sigma}_1^2 = (1+1)/2 = 1$.

(b) $\hat{\sigma}^2 = (3 \cdot 2/3 + 2 \cdot 1)/5 = 4/5 = 0.8$.

(c) $s_0(4) = -\frac{4}{2 \cdot 2/3} - \frac{1}{2}\ln\frac{2}{3} + \ln 0.6 = -3 + 0.203 - 0.511 = -3.308$. $s_1(4) = -\frac{4}{2} - 0 + \ln 0.4 = -2 - 0.916 = -2.916$. Class 1. Posterior $\sigma(s_1 - s_0) = \sigma(0.392) = 0.597$.

(d) $s_0(4) = 4 \cdot 2/0.8 - 4/1.6 + \ln 0.6 = 10 - 2.5 - 0.511 = 6.989$; $s_1(4) = 24/0.8 - 36/1.6 + \ln 0.4 = 30 - 22.5 - 0.916 = 6.584$. Class 0. Boundary: $s_1 - s_0 = 5x - 20 - 0.405 = 0 \Rightarrow x = 4.08$: $x = 4$ sits just on the class-0 side.

(e) QDA gives class 1 the larger variance (1 against 2/3), so class 1's density decays more slowly and still has mass at distance 2 from its mean, while class 0's tighter Gaussian has already dropped off at distance 2. LDA forces one spread on both and lets the prior and the midpoint 4.08 decide.

(f) LDA: 1 prior $+2 \cdot 10$ means $+55$ covariance $= 76$. QDA: $1 + 20 + 2 \cdot 55 = 131$. With $N = 60$ (30 per class) a 10×10 covariance per class is estimated from 30 points: singular or wildly noisy. Fit LDA (or naive Bayes); QDA needs far more data per class.

C3. Naive Bayes spam filter

(a) $\hat{\pi}_{\text{spam}} = 20/80 = 0.25$, $\hat{\pi}_{\text{ham}} = 0.75$. Denominator $18 + 1 \cdot 4 = 22$: spam $(11, 2, 7, 2)/22$; ham $(3, 10, 1, 8)/22$.

(b) m_1: free $\times 2$, win $\times 1$. $s_{\text{spam}} = \ln 0.25 + 2\ln\frac{11}{22} + \ln\frac{7}{22} = -1.386 - 1.386 - 1.145 = -3.918$. $s_{\text{ham}} = \ln 0.75 + 2\ln\frac{3}{22} + \ln\frac{1}{22} = -0.288 - 3.985 - 3.091 = -7.364$. Spam. Posterior $\sigma(-3.918 + 7.364) = \sigma(3.446) = 0.969$.

(c) m_2: meeting $\times 1$, report $\times 2$. $s_{\text{spam}} = -1.386 + \ln\frac{2}{22} + 2\ln\frac{2}{22} = -1.386 - 3 \cdot 2.398 = -8.580$. $s_{\text{ham}} = -0.288 + \ln\frac{10}{22} + 2\ln\frac{8}{22} = -0.288 - 0.788 - 2.023 = -3.099$. Ham, with posterior $\sigma(5.48) \approx 0.996$.

(d) With $\alpha = 0$, $\hat{\theta}_{\text{ham, win}} = 0/18 = 0$, so $s_{\text{ham}} = -\infty$ and the posterior of spam is exactly 1. One word never seen in 18 ham tokens vetoes ham with certainty: an artefact of a tiny training count, not evidence. Smoothing keeps every probability positive.

(e) $p(x \mid y = c) = \prod_j p(x_j \mid y = c)$. Words co-occur ("free" and "win" travel together), so the evidence is double-counted and the posteriors are overconfident (0.969 is too sure). But the *ranking* of the classes is often still right, and the decision only needs the argmax.

C4. Imbalance and prior shift

(a) With 2% positives the fraud region of the posterior is learned from few examples: high variance for the minority class. Subsampling gives the learner equal attention to both classes, which stabilises the estimate of the *boundary*; it breaks the *prior*: the model's probabilities are calibrated for a world where fraud is 50%.

(b) $\text{odds}_t = 0.8/0.2 = 4$. Prior-odds ratio $= (0.02/0.98)/(0.5/0.5) = 0.0204$. $\text{odds}_d = 4 \times 0.0204 = 0.0816$, so $r_d = 0.0816/1.0816 = 0.0755$.

(c) $t = 1/(1+20) = 0.0476$. Before correction $0.8 > t$: flag. After correction $0.0755 > 0.0476$: still flag, but now because misses are expensive, not because the model "thinks" 80%. Had the costs been equal ($t = 0.5$) the corrected decision would flip to "legitimate".

(d) Replace $\hat{\pi}$ by the deployment priors $(0.02, 0.98)$ in Bayes' rule. The class-conditionals

$p_{\hat{\theta}}(x \mid y)$ are unchanged: they were estimated per class and never depended on the mix.

(e) Precision $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$ depends on the prior. With 2% positives there are 49 legitimate transactions for every fraud, so the same false-alarm *rate* produces 49× more false alarms per true positive than on the balanced set. Recall and specificity transfer; precision does not.

C5. Choosing a route

(a) Generative: multinomial naive Bayes with smoothing. 300 examples against 5,000 features is far too little for a discriminative model to estimate 15,000 weights; the independence structure gives low-variance estimates and works well for word counts.

(b) Discriminative: a deep convolutional network trained with cross-entropy. Data are abundant, the boundary is complex and no credible $p(x \mid y)$ for photos exists; a misspecified generative model would converge to the wrong boundary.

(c) Combination: a discriminative classifier for the decision plus a generative or density model of $p(x \mid c)$ for a likelihood-threshold reject option; softmax confidence alone cannot detect unfamiliar inputs.

(d) Generative (or a calibrated discriminative model with the odds correction). With a generative model the fix is to swap $\hat{\pi}$ each season; the class-conditionals stay. Nothing needs retraining.

(e) Both give $s_c(x)$ affine in x. LDA fits means and a shared covariance by maximum likelihood on $p(x \mid y)$ and derives the boundary; logistic regression fits the boundary directly by conditional likelihood. LDA wins with little data when the classes are roughly Gaussian with equal covariance; logistic regression wins with more data or when the Gaussian assumption is wrong (outliers, skewed features), since it never has to model x.